

PART 4 — LEVEL 4 (H.C. Verma Level / JEE Advanced) ~120 Questions

SECTION 1: Crystal Lattices & Unit Cells (Q1–40)

1. A metal crystallizes in FCC structure. Atomic radius = $1.43 \text{ \AA}$. Calculate edge length and packing efficiency.
2. Calculate the density of copper (FCC) given atomic mass = 63.5 g/mol , atomic radius = $1.28 \text{ \AA}$.
3. BCC unit cell has edge $3 \text{ \AA}$. Find atomic radius and packing efficiency.
4. HCP metal has 2 atoms/unit cell, $c/a = 1.633$, atomic radius = $1.43 \text{ \AA}$. Calculate density.
5. Number of atoms per unit cell in FCC, BCC, SC, HCP and show relation to coordination number.
6. Edge length of NaCl unit cell = $5.64 \text{ \AA}$. Calculate distance between nearest ions.
7. CsCl structure: edge length = $4.12 \text{ \AA}$. Calculate radius of Cs^+ if $r_{\text{Cl}^-} = 1.67 \text{ \AA}$.
8. Compare packing efficiency of SC, BCC, FCC, HCP.
9. Calculate fraction of unoccupied volume in FCC and BCC.
10. A metal has density 8 g/cm^3 , atomic mass = 55.85 g/mol . Determine if it is BCC or FCC if atomic radius = $1.26 \text{ \AA}$.
11. Volume of unit cell of FCC metal with edge length $4 \text{ \AA}$.
12. Number of tetrahedral voids in FCC unit cell = ? Calculate fractional volume of voids.
13. Number of octahedral voids in FCC unit cell = ? Calculate fractional volume of voids.
14. A SC lattice contains atoms of radius $1 \text{ \AA}$. Calculate edge length, coordination number, and packing efficiency.
15. Given lattice constant a of FCC metal, find nearest neighbor distance.
16. Determine relation between atomic radius and edge length for BCC and FCC.
17. Calculate number of atoms along face diagonal of FCC unit cell.
18. Calculate number of atoms along body diagonal of BCC unit cell.
19. FCC lattice: edge length = $4 \text{ \AA}$. Atomic mass = 64 g/mol . Compute density.
20. BCC lattice: edge length = $3 \text{ \AA}$, atomic mass = 55.85 g/mol . Compute density.
21. Calculate edge length of FCC metal if density = 8.96 g/cm^3 , atomic mass = 63.5 g/mol .
22. SC lattice: edge length = $3 \text{ \AA}$, atomic mass = 50 g/mol . Compute density.
23. Compare coordination numbers of SC, BCC, FCC, HCP.
24. Explain why FCC and HCP have same packing efficiency.
25. For BCC: radius = $1.5 \text{ \AA}$. Compute edge length and packing fraction.
26. Compute volume fraction of atoms in BCC and SC lattices.
27. Calculate radius of atoms in CsCl structure from lattice constant.
28. Compute number of atoms/unit cell in HCP and FCC, compare.
29. Determine the type of cubic lattice from given density, atomic radius, and atomic mass.
30. FCC metal: lattice constant $4 \text{ \AA}$. Find mass of atoms per unit cell.
31. Calculate the distance between nearest neighbors in FCC lattice.
32. Compare packing efficiency of SC, BCC, FCC numerically.
33. Calculate volume of octahedral void in FCC lattice of edge $4 \text{ \AA}$.

34. Calculate volume of tetrahedral void in FCC lattice.
35. Derive formula for number of tetrahedral and octahedral voids in FCC lattice.
36. Edge length of NaCl unit cell = 5.64 Å. Calculate density.
37. Edge length of CsCl unit cell = 4.12 Å. Calculate density.
38. Explain why SC lattice is rarely observed in metals.
39. Compare nearest neighbor distances in BCC and FCC for same atomic radius.
40. Calculate total number of atoms in 1 cm³ of FCC copper (atomic mass = 63.5 g/mol, density = 8.96 g/cm³).

◆ SECTION 2: Imperfections in Solids (Q41–70)

41. Calculate fraction of Schottky defects in NaCl at 1000 K, formation energy = 2 eV.
42. Compute number of Frenkel defects in AgCl at 800 K, formation energy = 1 eV.
43. Calculate decrease in density of NaCl due to Schottky defect (fraction = 10⁻⁶).
44. Calculate number of F-centres formed in NaCl doped with alkali metal.
45. Explain why Frenkel defect does not change density.
46. Compute fraction of vacancies at given temperature using Boltzmann distribution.
47. Determine effect of Schottky defect on electrical conductivity.
48. Calculate vacancy concentration in ionic crystal at given temperature.
49. Compare Schottky and Frenkel defects with examples.
50. Edge dislocation: compute Burgers vector for given lattice.
51. Screw dislocation: determine displacement in crystal.
52. Compute strain energy associated with edge dislocation.
53. Calculate number of vacancies in NaCl at 1200 K.
54. Explain F-centres causing colour in alkali halides.
55. Compute fractional decrease in density due to Schottky defects in KCl.
56. Calculate activation energy for vacancy formation if fraction of defects is known at two temperatures.
57. Determine effect of Frenkel defect on conductivity in AgCl.
58. Compare Schottky and Frenkel defects quantitatively.
59. Compute change in lattice energy due to defects.
60. Explain why F-centres produce specific colours.
61. Edge dislocation: determine stress field around dislocation.
62. Compute dislocation density if given length of dislocations per unit volume.
63. Explain effect of dislocations on mechanical strength.
64. Compute energy per unit length of dislocation.
65. Determine number of vacancies at given temperature for given crystal.
66. Compare formation energies of Schottky and Frenkel defects.
67. Calculate change in lattice parameter due to defects.
68. Determine defect concentration in doped ionic crystal.
69. Compute electrical conductivity increase due to defects.
70. Explain role of defects in diffusion processes.

◆ SECTION 3: Electrical & Magnetic Properties (Q71–100)

71. Compute intrinsic carrier concentration in semiconductor at 300 K given band gap.
72. Determine conductivity of N-type semiconductor given doping level and mobility.
73. Compute mobility of charge carriers from resistivity.
74. Determine type of semiconductor using Hall coefficient.
75. Calculate Hall voltage for given current and magnetic field.
76. Compute susceptibility of paramagnetic substance given atomic magnetic moment.
77. Determine critical temperature for ferromagnet given magnetization data.
78. Compute magnetic moment per atom in ferromagnetic solid.
79. Calculate density of states for semiconductor at given energy.
80. Determine resistivity change with temperature for metal.
81. Compute carrier concentration in extrinsic semiconductor at room temperature.
82. Determine Curie temperature for ferromagnet from experimental data.
83. Compute Meissner effect parameters in superconducting sample.
84. Calculate penetration depth in superconductor.
85. Determine magnetic field inside paramagnetic material.
86. Compute magnetization for given magnetic field.
87. Calculate Hall coefficient for metal given carrier density.
88. Determine effect of doping on conductivity numerically.
89. Compute carrier drift velocity given current and area.
90. Determine type of semiconductor from sign of Hall coefficient.
91. Compute susceptibility of diamagnetic material given orbital data.
92. Calculate critical current in superconductor.
93. Determine energy gap from temperature dependence of conductivity.
94. Compute change in resistivity with temperature.
95. Calculate magnetization for ferromagnetic material using Curie law.
96. Determine effect of alloying on conductivity.
97. Compute electron mobility in semiconductor.
98. Determine carrier concentration using Hall measurements.
99. Compute penetration depth and critical field in superconductor.
100. Explain numerical effect of defects on electrical conductivity.

◆ SECTION 4: High-Level Numericals & Conceptual Problems (Q101–120)

101. Calculate total number of atoms in 1 cm^3 of FCC metal given density and atomic mass.
102. Determine lattice type from given density, atomic radius, and atomic mass.
103. Compute energy change due to formation of Schottky defects at high temperature.
104. Compare packing efficiency of HCP vs FCC using numerical calculation.
105. Compute edge length of unit cell given atomic radius for FCC/BCC.
106. Determine carrier concentration in intrinsic semiconductor numerically.

107. Compute volume fraction of tetrahedral and octahedral voids in FCC lattice.
108. Determine type of crystal structure from density and atomic mass.
109. Compute effect of Frenkel defect on conductivity numerically.
110. Calculate Hall voltage for given semiconductor sample.
111. Determine atomic radius in CsCl given lattice constant.
112. Compute energy associated with dislocations in metal.
113. Calculate fractional change in density due to defects.
114. Determine type of semiconductor from Hall coefficient sign and magnitude.
115. Compute nearest neighbor distance in BCC and FCC.
116. Determine number of vacancies at given temperature using Arrhenius equation.
117. Calculate carrier mobility from conductivity and carrier concentration.
118. Determine packing fraction for given HCP structure numerically.
119. Compute total number of F-centres for a doped ionic crystal.
120. High-level conceptual problem: Compare mechanical, electrical, and magnetic properties for given solid with defects and crystal type.